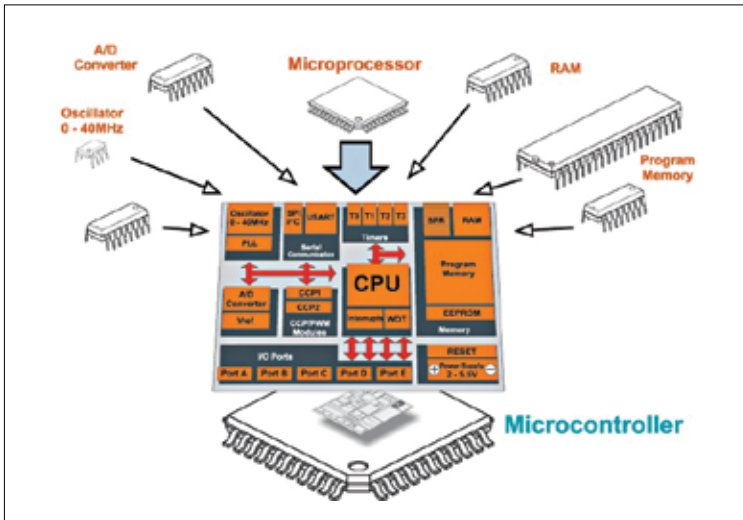




Comparing Microprocessor and Microcontroller

RAM, ROM etc. integrated on the chip. All these components and other peripherals need to be added externally to make a working system. A microcontroller, on the other hand, has a CPU, RAM, ROM etc. all put together on a single IC (integrated chip).

- **Application:** Microprocessors are used in desktop computers and laptop, whereas microcontrollers are used in products where interfacing with hardware is necessary such as in robots and MP3 players.
- **Cost:** So where does Arduino™ lie on the microcontroller landscape?

Arduino™ is a microcontroller-based board which makes possible the easier development of hardware to develop quick prototypes. The microcontrollers used are either 8-bit Atmel AVR series microcontrollers or 32-bit Atmel ARM-based SMART series microcontrollers. They can be programmed using a personal computer. Every Arduino™ board has digital and analog pins, power pins, programming headers, serial communication, Pulse Width Modulation (PWM, explained more in chapter 6 section Voltage Control using PWM) and Inter IC connect protocol (I2C) support.

What's in it for me?

Being an Open Source project, the design files of every Arduino™ board are available online for anyone to modify and use for free.